

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EE) Examination

Nov-Dec 2012

EE207/EE207.01 Network Analysis

Date: 27.11.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q – 1 Explain KCL and KVL using suitable examples. 07
- Q – 2 (a) State and explain (i) Thevenin's theorem and (ii) Norton's theorem in brief giving suitable examples. 06
- (b) In the Wheatstone's bridge circuit shown in Figure 1, a galvanometer of 10 ohm is connected across BD. Using Thevenin's theorem, calculate the current through the galvanometer. 06

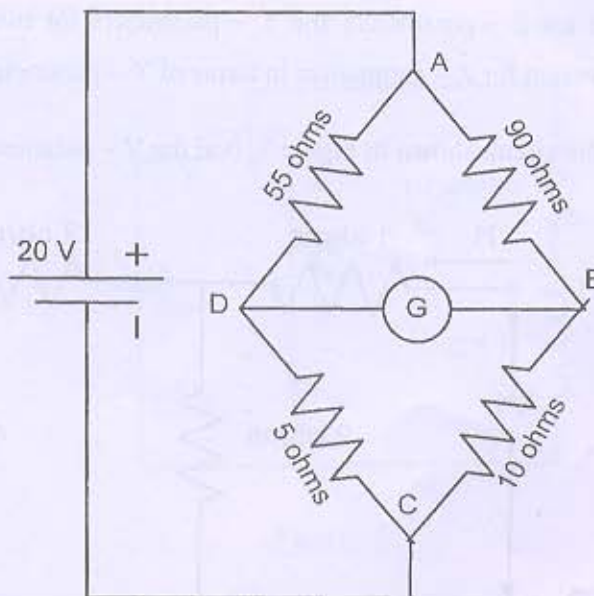


Figure 1

- (c) How inductor and capacitor will behave at $t = 0$ Sec and $t = \infty$ Sec when switched to DC source. Draw equivalent networks. 02

OR

- Q-2 (a) State and explain the maximum power transfer theorem. Derive the condition for maximum power transfer to the load for DC circuits. 06
- (b) In the network shown in Figure 2, find the current through the 10 ohm resistance using superposition theorem. 06

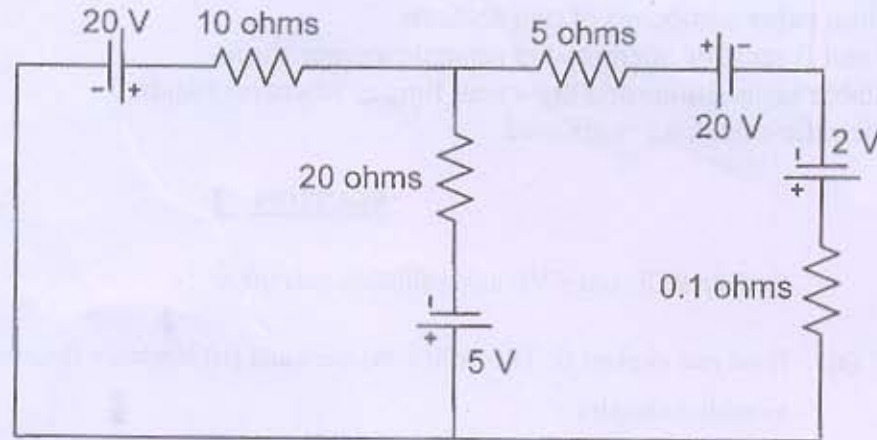


Figure 2

- (c) Define the following terms (i) Linear (ii) Bilateral (iii) branch (iv) Time invariant. 02
- Q-3 (a) What are Z - parameters and Y - parameters for two port network? Derive the expression for Z - parameters in terms of Y - parameters and vice versa. 07
- (b) For the circuit shown in Figure 3, find the Y - parameters. 07

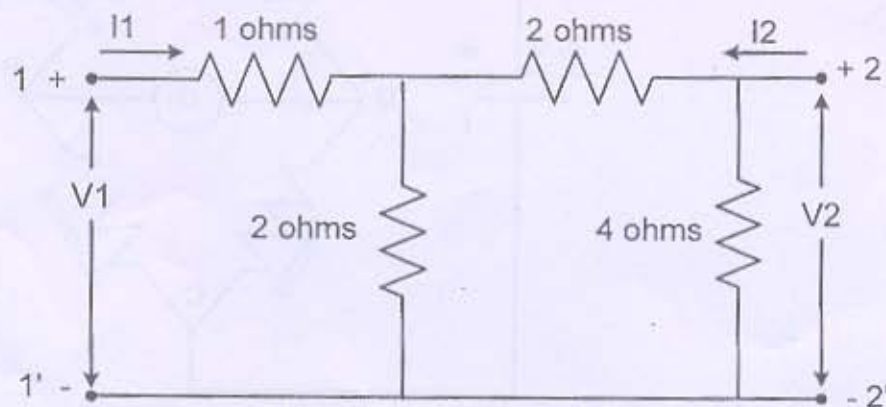


Figure 3

OR

- Q-3 (a) Explain about the h - parameters of two port network. Where do we find its application? 07

(b) For the circuit shown in Figure 4, find the h -parameters. 07

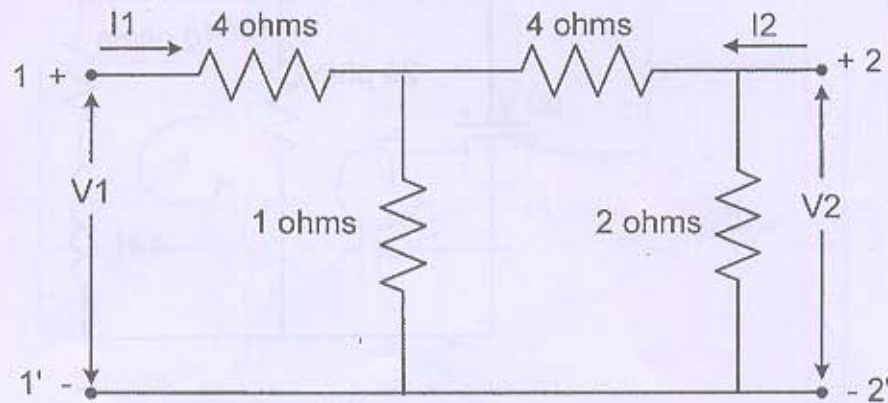


Figure 4

SECTION - II

Q-4 What is duality? Prepare a list of dual quantities encountered in electrical engineering. Describe the procedure to draw dual of network. 07

Q-5 (a) Derive the step response of series R-C circuit having DC excitation. 07

(b) In the circuit shown in Figure 5, determine the current $i(t)$ when the switch is changed from position 1 to 2 at time $t = 0$. 07

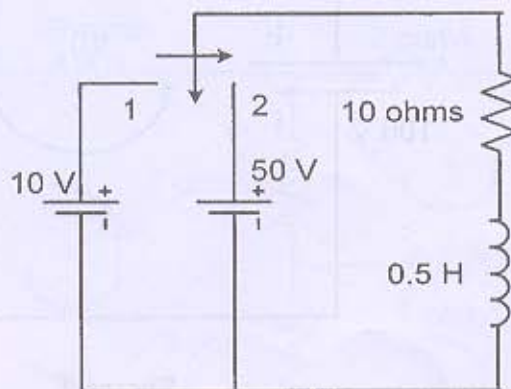


Figure 5

OR

Q-5 (a) For the circuit shown in Figure 6, obtain the equations for $i_1(t)$ and $i_2(t)$ when the switch is closed at $t = 0$. 07

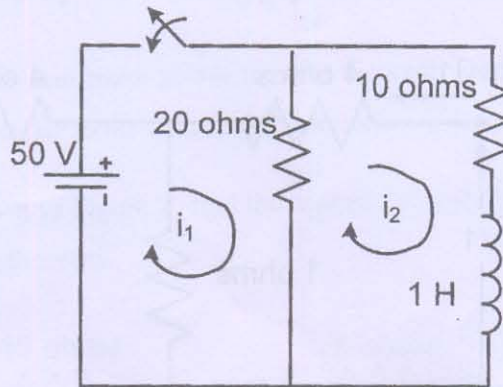


Figure 6

- (b) In the series R-C circuit shown in Figure 7, the capacitor has an initial charge $q_0 = 2500 \mu\text{C}$. At $t = 0$, the switch is closed and the constant voltage source $V = 100$ Volts is applied to the circuit. Use the method of Laplace Transform to find out the current. 07

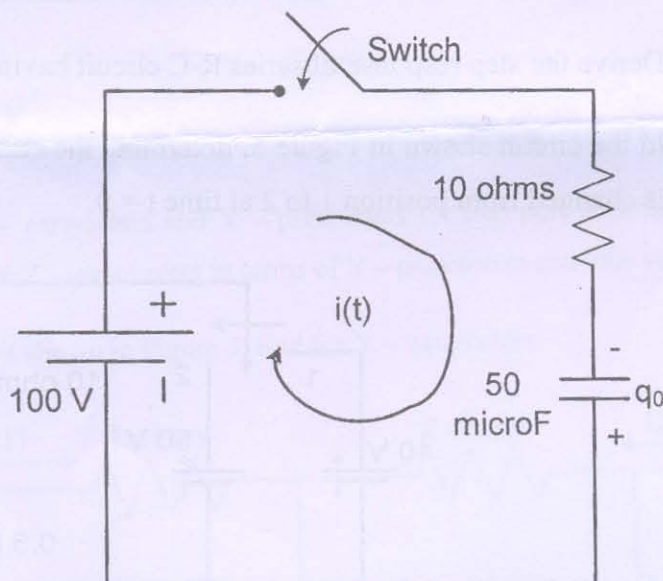


Figure 7

- Q-6 (a) Explain the procedure to find out incidence matrix by giving a suitable example. 07
- (b) For the system shown in Figure 8, draw the oriented graph and write (i) incidence matrix and (ii) tie set matrix 07

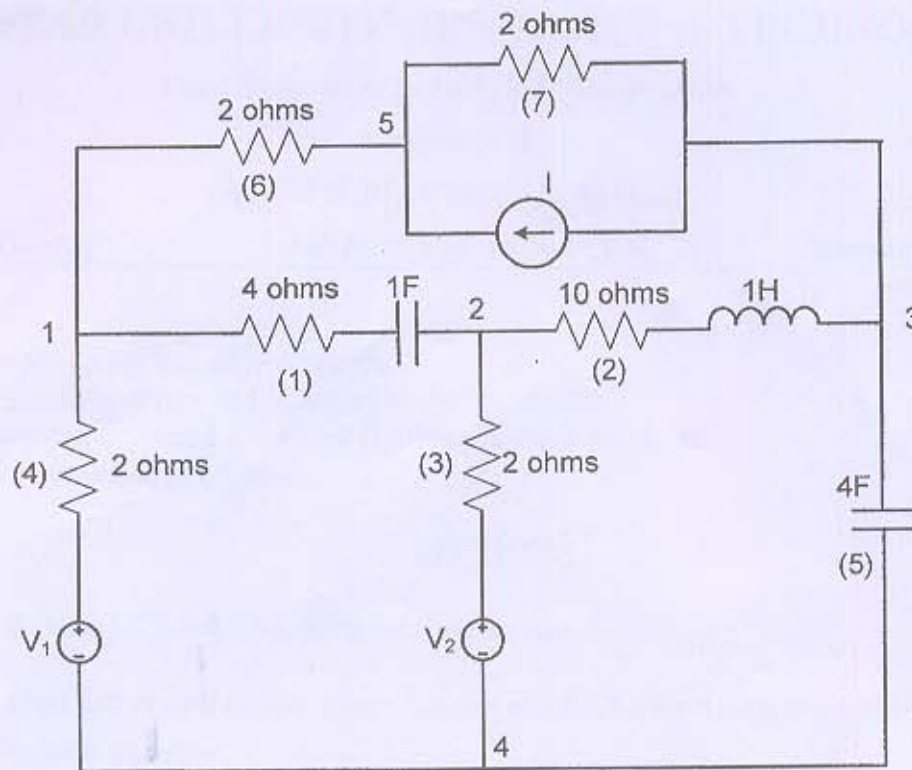


Figure 8

OR

- (b) Determine the mesh current I_1 in the circuit shown in Figure 9.

07

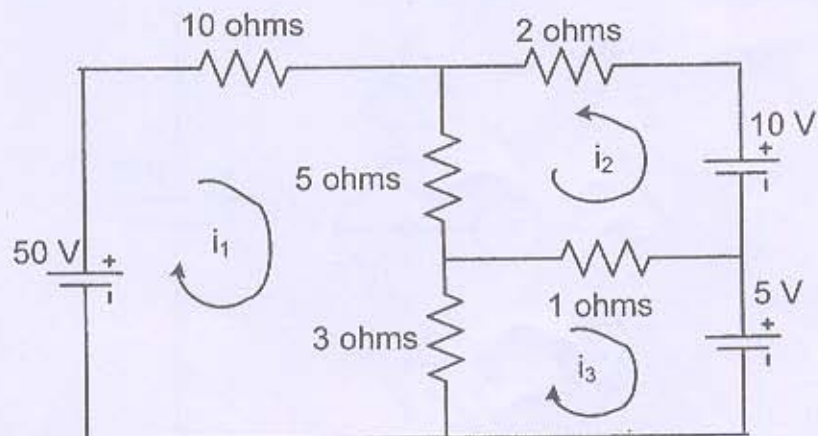


Figure 9
